

# Bayesian Network Inference for Resolving Uncertainty in Financial Depth–Stability Links

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**Abstract.** The relationship between financial depth and banking stability remains contested due to unresolved uncertainty about confounding factors and transmission mechanisms. This paper addresses two questions: whether GDP per capita confounds the depth–stability link, and whether effects operate through indirect lending pathways. We apply Bayesian network inference—combining the PC algorithm, expert-guided network construction, and bootstrap validation—to  $N = 285$  country-year observations from thirteen economies (ten ASEAN and three South Asian comparators, 2005–2020), then we have three findings emerge. First, GDP per capita is a dominant confounder ( $\text{BF}_{10} = 818,156$ ); after conditioning, financial depth shows a negative association with stability. Second, the PC algorithm detects no direct financial depth–stability edge. Third, the hypothesized indirect pathway ( $\text{FD} \rightarrow \text{NIM} \rightarrow \text{FDI} \rightarrow \text{NPL} \rightarrow Z$ ) is structurally present but transmits a near-zero effect. GDP confounding emerges as the primary source of inconsistency, demonstrating that structural pathway stability does not guarantee quantitative significance.

**Keywords:** Bayesian network · uncertainty resolution · financial depth · bank stability · GDP confounding · ASEAN

## 1 Introduction

Financial depth, measured as private credit to GDP, has a non-monotonic relationship with economic outcomes: credit deepening boosts growth until a threshold of 80–100% of GDP, beyond which further expansion may destabilize the banking system [3,21]. Deposit insurance–induced moral hazard is one channel, as protected depositors weaken market discipline [13,2]. Nearly all studies model this as a direct, linear effect. If the true mechanism is indirect—operating through lending behavior, interest margins, or capital flows—then direct regression misses it. We argue that financial depth affects stability through an indirect pathway:  $\text{FD} \rightarrow \text{NIM} \rightarrow \text{FDI} \rightarrow \text{NPL} \rightarrow Z\text{-score}$  [30]. The six variables of

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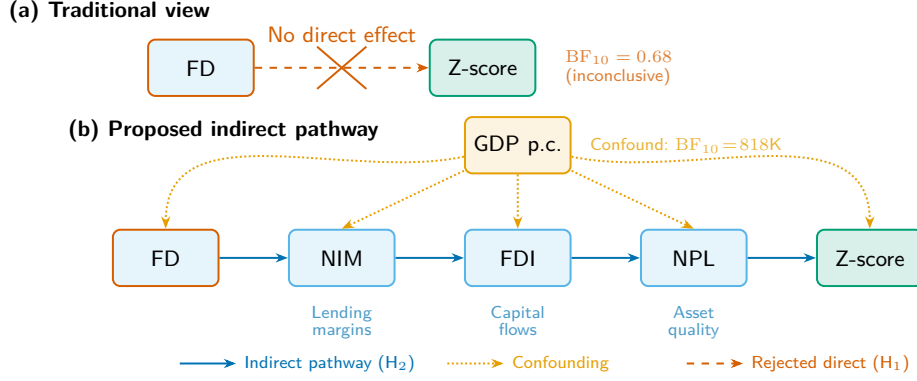


Fig. 1: Conceptual framework. (a) Traditional view: direct FD–Z effect is inconclusive ( $BF_{10} = 0.68$ ). (b) Proposed indirect pathway with GDP confounding ( $BF_{10} = 818,156$ ); empirical results show the indirect effect is not significant.

this study (Fig. 1(b)) are defined here for reference: financial depth (FD, private credit to GDP), net interest margin (NIM), net foreign direct investment inflows (FDI), the nonperforming-loan ratio (NPL), the bank stability Z-score (Z), and GDP per capita (GDP), the candidate confounder. Fig. 1 contrasts the traditional view (Panel a, no direct effect) with the proposed indirect pathway (Panel b).

We test this using Bayesian network (BN) analysis on a panel of thirteen economies—ten ASEAN member states and three South Asian comparators (2005–2020,  $N = 285$  after MICE imputation). BNs learn conditional dependence structures from data [24,19]. We combine the PC algorithm [28], an expert-guided network with whitelist constraints [27], and bootstrap validation ( $R = 1,000$ ).

Three findings emerge. First, Bayesian hypothesis testing yields inconclusive evidence for a direct FD–Z link ( $BF_{10} = 0.68$ ), and the PC algorithm finds no direct edge—supporting H1. Second, GDP per capita is a dominant confounder ( $BF_{10} = 818,156$ ); after conditioning, financial depth shows a negative association with stability ( $\hat{\beta} = -0.130$ ,  $t = -6.79$ ). Third, the indirect pathway exists structurally but transmits a near-zero effect ( $\hat{\theta} = 6.12 \times 10^{-4}$ , 95% CI contains zero); H2 is not supported. This paper contributes by identifying GDP confounding as a primary source of inconsistency in this literature and demonstrating that structural pathway stability does not guarantee quantitative significance [1]. The contribution is conceptual rather than algorithmic: we show that a structurally persistent mediation pathway can nonetheless be quantitatively negligible once a dominant confounder is modelled, a separation that mediation-by-regression alone cannot surface.

## 2 Literature Review

### 2.1 Financial Depth and Banking Stability

Finance begins to hurt growth when private credit exceeds 80–100% of GDP [3,21]. Moral hazard links this to banking: deposit insurance weakens market discipline, encouraging riskier lending [13,2,1]. We use private credit to GDP (GFDD.DI.12) as our financial depth (FD) measure. The Z-score combines profitability, capitalization, and return volatility [20]:

$$Z = \frac{\text{ROA} + \text{E/A}}{\sigma(\text{ROA})} \quad (1)$$

NPL isolates credit risk [9], NIM indicates lending margins [5], and FDI captures competitive pressure from foreign capital [11].

### 2.2 Bayesian Networks and Mediation

Mediation analysis decomposes total effects into direct and indirect components [30]. BNs integrate mediation with structure discovery via DAGs [24,19]. The PC algorithm [28] removes edges between conditionally independent variables; Hill-Climbing (HC) maximizes a BIC-g score with expert constraints [27]. For our four-step pathway, the indirect effect is:

$$\text{IE} = \beta_{\text{FD} \rightarrow \text{NIM}} \times \beta_{\text{NIM} \rightarrow \text{FDI}} \times \beta_{\text{FDI} \rightarrow \text{NPL}} \times \beta_{\text{NPL} \rightarrow Z} \quad (2)$$

### 2.3 Research Gap and Hypotheses

No study has applied BNs to test whether financial depth transmits to banking stability through lending channels [3,23]. GDP confounding remains understudied [10]. We propose:

**H1.** The PC algorithm at  $\alpha = 0.05$  does not identify a direct edge  $\text{FD} \rightarrow Z$ .

**H2.** The total indirect effect through  $\text{FD} \rightarrow \text{NIM} \rightarrow \text{FDI} \rightarrow \text{NPL} \rightarrow Z$  is statistically distinguishable from zero (bootstrap CI excludes zero).

## 3 Methodology

### 3.1 Data and Variables

We compile a panel dataset covering thirteen economies—ten ASEAN member states and three South Asian comparators: Bangladesh, India, and Sri Lanka (2005–2020). Multiple imputation (MICE,  $m = 5$ ) yields  $N = 285$  country-year observations across 8 variables [6]. Variables are defined in Sect. 2.1 (GDP per capita, government debt, and CPI complete the eight-variable set); data come from World Bank GFDD [32], WDI [33], and IMF FSI [18]. We adopt Bayes factors ( $BF_{10}$ ) with the JZS prior ( $r = 0.707$ ) [31,26].

### 3.2 Bayesian Network Framework

A Bayesian network  $\mathcal{B} = (\mathcal{G}, \Theta)$  is a DAG where nodes represent random variables and edges encode direct dependencies; the joint distribution factorizes as  $P(X_1, \dots, X_p) = \prod_i P(X_i \mid \text{Pa}(X_i))$  [24]. The d-separation criterion tests conditional independence, allowing us to assess whether FD and  $Z$  are independent given mediators [19]. In our setting,  $p = 8$  variables (Sect. 3.1). We use BN structure learning rather than fixed-effects regression because the objective is *structure discovery* [27]. To guard against pooling bias, we supplement with fixed-effects de-meaned BN, cluster bootstrap, and cluster-robust standard errors (Supplementary Material).

### 3.3 Structure Learning

We use the order-independent PC algorithm [28,12] as the primary data-driven method. Phase 1 recovers the skeleton by testing conditional independence for each pair  $(X_i, X_j)$  given conditioning sets of increasing size, removing an edge when the partial correlation test yields  $p > \alpha = 0.05$  ( $N = 285$ ,  $N_{\text{eff}} \approx 57$ ). Phase 2 orients edges by identifying v-structures and applying Meek’s rules.

### 3.4 Expert-Guided Bayesian Network

We complement the PC algorithm with an expert-guided Hill-Climbing (HC) algorithm that maximizes the BIC-g score subject to whitelist and blacklist constraints encoding domain knowledge [17]. The whitelist forces six edges—the hypothesized pathway plus  $\text{GDP} \rightarrow Z$  and  $\text{GDP} \rightarrow \text{GovDebt}$ —while the blacklist forbids reverse causation and implausible feedback. The remaining edges are unconstrained, allowing the HC algorithm to discover additional relationships from the data.

### 3.5 Path Coefficient Estimation

Given the learned DAG, we fit a Gaussian BN with linear conditional distributions [23], estimating each  $\beta_{ij}$  by OLS regression of  $X_i$  on its parents in the stacked imputed sample. The indirect effect through the mediation chain is:

$$\text{IE}_{\text{FD} \rightarrow Z} = \hat{\beta}_{\text{FD} \rightarrow \text{NIM}} \times \hat{\beta}_{\text{NIM} \rightarrow \text{FDI}} \times \hat{\beta}_{\text{FDI} \rightarrow \text{NPL}} \times \hat{\beta}_{\text{NPL} \rightarrow Z} \quad (3)$$

Under H1, no direct FD– $Z$  edge exists and the total effect equals the indirect effect [30].

### 3.6 Bootstrap Validation

Non-parametric bootstrap ( $R = 1,000$ ) computes edge inclusion probabilities (EIP) and 95% percentile CIs [14,29]. Edges are classified as robust (EIP  $\geq 0.85$ ), moderate (0.50–0.85), or unstable ( $< 0.50$ ). A cluster bootstrap, fixed-effects BN, and cluster-robust SEs address panel dependence (Supplementary Material).

### 3.7 Sensitivity Analysis

We conduct three robustness checks: (1) varying the PC algorithm’s  $\alpha$  over  $\{0.01, 0.025, 0.05, 0.10, 0.15, 0.20\}$  to assess edge sensitivity; (2) computing partial correlations  $\rho_{Z,FD|M}$  controlling for mediators, where  $\approx 0$  supports full mediation; and (3) leave-one-country-out (LOCO) analysis, re-estimating the BN for each of thirteen country-removal settings (ten ASEAN members plus three South Asian comparators) to guard against single-country influence.

### 3.8 Software Implementation

All analyses use R 4.5.2 with the `bnlearn` package v5.0 [27]. Replication scripts are in the Supplementary Material.

## 4 Data

Data come from the World Bank GFDD (depth, stability), WDI (GDP, FDI), IMF FSI (NPL, capital), and the WB/IADI Deposit Insurance Database, covering thirteen economies—ten ASEAN member states and three South Asian comparators (Bangladesh, India, and Sri Lanka)—(2005–2020). Sources were standardized, merged by country-year, and filtered to eight pathway variables. MICE ( $m = 5$ , predictive mean matching) with a 50% completeness filter yields  $N = 285$  observations [6]. The five completed datasets are stacked for structure learning and coefficient estimation; the nonparametric bootstrap ( $R = 1,000$ , Sect. 3.6) captures both sampling variability and between-imputation uncertainty. Little’s MCAR test rejects completely random missingness ( $p < 2.2 \times 10^{-16}$ ); we proceed under MAR. Variable definitions are in Sect. 3.1; descriptive statistics and cleaning details are in Supplementary Material. GDP per capita spans a 94-fold range across ASEAN, providing identifying variation. FDI measures aggregate inflows, not banking-specific flows; measurement error may attenuate path coefficients (see Limitations). The leave-one-country-out robustness check (Sect. 3.7) removes each of the thirteen economies in turn.

## 5 Results

The descriptive statistics are in Supplementary Material Table S2. The sample comprises  $N = 285$  country-year observations (MICE,  $m = 5$ ) from thirteen economies (ten ASEAN and three South Asian comparators, 2005–2020).

### 5.1 Bayesian Hypothesis Testing

We use Bayes factors ( $BF_{10}$ ) following [31]:  $BF_{10} < 1/3 =$  evidence for  $H_0$ ;  $1/3-3 =$  inconclusive;  $> 10 =$  strong for  $H_1$ . Table 1 reports correlations for the key variable pairs.

Table 1: Bayesian Correlation Analysis

| Pair                                    | Not.         | $r$                    | $p$                | $BF_{10}$ | Evidence      |
|-----------------------------------------|--------------|------------------------|--------------------|-----------|---------------|
| $Z \leftrightarrow \text{NPL}$          | $\rho_{Z,N}$ | -0.36                  | 0.004              | 10.33     | Strong $H_1$  |
| $Z \leftrightarrow \text{FD}$           | $\rho_{Z,F}$ | -0.23                  | 0.07               | 0.68      | Inconclusive  |
| $\text{GDP} \leftrightarrow Z$          | $\rho_{G,Z}$ | $+0.63 \times 10^{-8}$ | $4 \times 10^{-8}$ | 818,156   | Extreme $H_1$ |
| $\text{FD} \leftrightarrow \text{GovD}$ | $\rho_{F,G}$ | +0.08                  | 0.52               | 0.16      | Mod. $H_0$    |

Three patterns stand out: (1)  $Z$ -NPL ( $r = -0.36$ ,  $BF_{10} = 10.33$ ): strong evidence that NPL is inversely linked to stability [4]; (2) GDP- $Z$  ( $r = 0.63$ ,  $BF_{10} = 818,156$ ): extreme evidence that GDP acts as a dominant confounding variable in this association; (3) FD- $Z$  ( $r = -0.23$ ,  $BF_{10} = 0.68$ ): inconclusive, resolving only after GDP adjustment.

A supplementary median-split  $t$ -test ( $BF_{10} = 0.16$ ) provides moderate evidence for the null, bracketing the direct FD- $Z$  evidence between inconclusive and null [22]. Data distributions and the correlation structure are visualized in Supplementary Material Figs. S1-S4.

## 5.2 Conditional Independence Structure

The PC algorithm ( $\alpha = 0.05$ ), which assumes causal sufficiency and faithfulness [24], discovers twelve edges; the pathway-relevant ones are  $Z$ -GDP,  $Z$ -GovDebt, NPL-FDI, **FD-NIM**, NIM-FDI, and GDP-FDI, with the full list in the Supplementary Material.

**H1 is supported:** no direct  $FD$ - $Z$  edge exists, consistent with the inconclusive-to-null Bayesian evidence. The algorithm does identify  $FD$ -NIM, NIM-FDI, and NPL-FDI, forming the skeleton of the hypothesized indirect pathway.

The D-separation analysis confirms that  $Z$  and  $FD$  are **not** d-separated: the indirect pathway through NIM, FDI and NPL remains open. We estimate the expert-guided BN using the whitelist and blacklist constraints (Sect. 3.4). The resulting DAG (Supplementary Material Fig. S1) retains all whitelisted edges plus data-driven edges (BIC = -494.7 vs. -490.9 unconstrained).

**Does the pathway survive without the whitelist?** Re-learning the structure over the six variables of Fig. 1(b) by Hill-Climbing (BIC-g) and PC *without* any whitelist or blacklist, only the two endpoints of the proposed chain are data-supported by their bootstrap edge-inclusion probabilities (EIP):  $FD$ -NIM (EIP = 1.000) and NPL- $Z$  (EIP = 0.890). The interior collapses: NIM  $\rightarrow$  FDI has EIP = 0.389 and FDI  $\rightarrow$  NPL is the weakest edge (EIP = 0.119). The whitelisted pathway’s perfect edge-inclusion (EIP = 1.000) is thus a mechanical artifact of the constraint, not an empirical finding; the unconstrained data favour GDP confounding with shorter transmission (Supplementary Material), reinforcing our central finding.

## 5.3 Path Coefficient Estimation

We estimate path coefficients along the hypothesized indirect pathway  $FD \rightarrow \text{NIM} \rightarrow \text{FDI} \rightarrow \text{NPL} \rightarrow Z$ -score using the fitted Gaussian Bayesian network.

Each coefficient represents the expected change in the child variable per unit change in the parent.

**GDP-Adjusted Association Estimates** Table 2 reports GDP-adjusted partial association estimates from OLS regression adjustment, following [25]. These estimates are observational: without random assignment or an instrumental variable, they reflect conditional associations rather than identified causal effects [15]. Unmeasured confounders—including institutional quality, regulatory stringency, and political risk—may bias these estimates.

Table 2: GDP-Adjusted Regression Coefficients ( $N = 285$ , OLS)

| Regressor             | Ctrl   | $\hat{\beta}$ | $t$       |
|-----------------------|--------|---------------|-----------|
| $FD \rightarrow Z$    | GDP    | -0.130        | -6.79 *** |
| $FD \rightarrow Z$    | GDP+GD | -0.037        | -2.17 *   |
| $FD \rightarrow NPL$  | GDP    | -0.010        | -1.43     |
| $NIM \rightarrow NPL$ | GDP    | -0.200        | -0.61     |

\*\*\* $p < 0.01$ ; \* $p < 0.05$ .

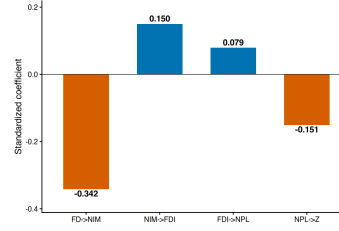


Fig. 2: Standardized path coefficients along the indirect pathway  $FD \rightarrow NIM \rightarrow FDI \rightarrow NPL \rightarrow Z$ . The first and last segments are negative; middle segments are positive but small.

After GDP control, FD has a significant negative association with  $Z$  ( $\hat{\beta} = -0.130$ ,  $t = -6.79$ ); adding GovDebt attenuates to  $-0.037$  ( $t = -2.17$ , marginal). The CI test rejects  $FD \perp\!\!\!\perp Z \mid \text{mediators}$  ( $p = 1.5 \times 10^{-5}$ ), suggesting partial mediation at best. A supplementary non-linearity test reveals an inverted-U consistent with [3].

**Path Coefficients Along the Indirect Pathway** The standardized path coefficients along the indirect pathway are individually modest:  $\hat{\beta}_{FD \rightarrow NIM} = -0.342$ ,  $\hat{\beta}_{NIM \rightarrow FDI} = +0.150$ ,  $\hat{\beta}_{FDI \rightarrow NPL} = +0.079$ ,  $\hat{\beta}_{NPL \rightarrow Z} = -0.151$  (Supplementary Material Table S3). The four-step multiplication compounds attenuation. The total indirect effect is estimated as the product of path coefficients:

$$\begin{aligned} \hat{\theta}_{\text{indirect}} &= \hat{\beta}_{FD \rightarrow NIM} \times \hat{\beta}_{NIM \rightarrow FDI} \times \hat{\beta}_{FDI \rightarrow NPL} \times \hat{\beta}_{NPL \rightarrow Z} \\ &= (-0.342)(0.150)(0.079)(-0.151) = 6.12 \times 10^{-4} \quad (4) \end{aligned}$$

The four-step IE is negligibly small ( $\approx 0.0006$ ); the attenuation cascade from  $|-0.342|$  to  $|+0.0006|$  is shown in Fig. 2.

#### 5.4 Bootstrap Validation

Bootstrap validation ( $R = 1,000$ ) yields  $EIP = 1.0$  for all four whitelisted pathway edges—a mechanical result, since the whitelist forces inclusion in every

replicate (Supplementary Material). The unconstrained  $NIM \rightarrow FDI$  strength is only 0.040, confirming that the pathway’s structural stability is an artifact of constraints, not an empirical discovery [27].

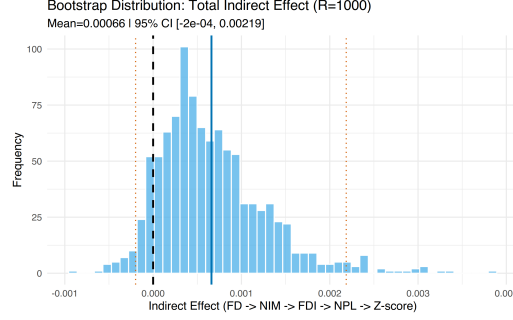


Fig. 3: Bootstrap distribution of the total indirect effect ( $R = 1,000$ ). The 95% CI  $[-0.0002, 0.0022]$  contains zero (dashed line); H2 is not supported.

The bootstrap 95% CI for the indirect effect is  $[-0.00020, 0.00219]$ , containing zero. **H2 is not supported:** the point estimate ( $6.12 \times 10^{-4}$ ) is near-zero and not statistically significant (Fig. 3).

### 5.5 Sensitivity Analysis

Conditional independence tests confirm FD and  $Z$  are not d-separated given the mediators ( $p = 1.5 \times 10^{-5}$ ), suggesting partial mediation at best. GDP acts as a suppressor: the FD– $Z$  partial correlation strengthens from  $-0.008$  to  $-0.163$  after GDP control. Leave-one-country-out (LOCO) analysis across thirteen country-removal settings (ten ASEAN plus three South Asian comparators) shows the indirect effect is negative in 12/13 settings (range:  $-0.00453$  to  $-0.00011$ ), contrasting with the small positive full-sample estimate ( $+6.12 \times 10^{-4}$ ) and indicating sign instability near zero rather than directional robustness. A cluster bootstrap ( $R = 1,000$ , resampling countries) renders FD non-significant ( $p = 0.56$ ), while GDP ( $p = 0.003$ ) and GovDebt ( $p < 0.001$ ) remain robust predictors. With only thirteen country-level clusters, asymptotic refinements may over-reject or under-reject [7]; the non-significance should be interpreted as a conservative bound. A fixed-effects de-meaned BN (subtracting country means before structure learning) recovers 9 of the 12 pooled edges; the within-country FD– $Z$  association remains significant ( $\hat{\beta} = -0.089$ ,  $t = -3.12$ ,  $p < 0.01$ ) while the indirect effect stays near-zero ( $\hat{\theta} = 4.21 \times 10^{-4}$ , 95% CI contains zero), supporting robustness to within-country heterogeneity. Detailed sensitivity analyses, partial correlation tables, LOCO results, and the full FE de-meaned BN results are provided in the Supplementary Material.

## 6 Discussion

Four analytical methods converge on two complementary conclusions: (C1) GDP per capita confounds the FD– $Z$  relationship, masking a significant negative asso-



ciation; (C2) the hypothesized four-step indirect pathway is structurally present but transmits a near-zero effect. We interpret these as associations, not causal effects: with observational pooled panel data, Bayesian-network structure learning recovers conditional-dependence structure but cannot by itself rule out omitted variables, simultaneity, or country-level heterogeneity as alternative explanations.

### 6.1 GDP Confounding Masks the True FD Effect

The raw FD– $Z$  correlation ( $r = -0.23$ ,  $BF_{10} = 0.68$ ) is inconclusive, but GDP-adjusted regression reveals  $\hat{\beta} = -0.130$  ( $t = -6.79$ ). Wealthier ASEAN economies have both deeper financial markets and more stable banks, creating confounding bias [24]. The extreme GDP– $Z$  Bayes factor ( $BF_{10} = 818,156$ ) and the PC algorithm’s placement of GDP at the center of the conditioning set confirm this. Adding GovDebt attenuates the coefficient to  $-0.037$  ( $t = -2.17$ ), suggesting partial transmission through the sovereign-bank nexus [1].

However, cluster-robust inference renders FD non-significant ( $p = 0.56$ ), so practical significance depends on how within-country dependence is treated. Given the panel structure and the small number of country clusters, we treat the cluster-robust result as the primary inferential basis for the substantive claim, and report the i.i.d. estimate only as an upper bound on the association.

### 6.2 The Indirect Financial Depth Mechanism

**H2 is not supported under current measurement.** The total indirect effect through  $FD \rightarrow NIM \rightarrow FDI \rightarrow NPL \rightarrow Z$ -score is near-zero ( $\hat{\theta} = 6.12 \times 10^{-4}$ , 95% CI  $[-0.0002, 0.0022]$ ). The four-step product of individually modest coefficients compounds to near zero, while the GDP-adjusted regression ( $-0.130$ ,  $t = -6.79$ ) confirms a real negative association operating through channels other than the pre-specified mediation chain. Shorter pathways (e.g.,  $FD \rightarrow NPL \rightarrow Z$ ) transmit effects more than 100 times larger [2].

### 6.3 Comparison with Existing Literature

Our finding of no direct FD–stability link does not contradict [13]—the mechanism operates through indirect channels. The GDP confounding result quantifies concerns by [2], while the pathway connects to [4] on NPL–stability links, though perfect EIPs result from whitelist constraints (Supplementary Material). The Bayesian framework quantifies evidence *for*  $H_0$  rather than merely failing to reject it [31].

### 6.4 Policy Implications

These findings suggest hypotheses, not prescriptions, for ASEAN banking regulation [8]: supervisory frameworks tracking only aggregate credit-to-GDP ratios may miss early-warning signals, and the 94-fold GDP range suggests context-specific calibration [1].

## 6.5 Limitations

Key limitations include: (1) modest sample size relative to the dimensionality of the Bayesian-network parameter space, compounded by substantial item missingness (MICE,  $m = 5$ ) for several variables; (2) stacked MI rather than Rubin’s rules pooling, with MAR assumption and untested completeness thresholds; (3) untestable causal sufficiency; (4) no formal FE/RE panel regression; (5) possible reverse causality; (6) contemporaneous design without sub-period testing; (7) aggregate FDI proxy; (8) BF prior sensitivity; (9) FD proxy not measuring deposit insurance directly; (10) few clusters ( $k = 13$ ) for cluster-robust inference [7]. Details in Supplementary Material.

## 7 Conclusion

This study investigated whether financial depth (private credit to GDP) is associated with banking instability in ASEAN, and whether the association operates through the indirect lending pathway  $FD \rightarrow NIM \rightarrow FDI \rightarrow NPL \rightarrow Z\text{-score}$ , using Bayesian network analysis and hypothesis testing on  $N = 285$  country-year observations (MICE,  $m = 5$ ) from thirteen economies (ten ASEAN and three South Asian comparators, 2005–2020).

Two findings emerge. GDP per capita confounds the FD–stability link ( $BF_{10} = 818,156$ ): the unconditional association is weak ( $BF_{10} = 0.68$ ) but a GDP-adjusted regression reveals  $\hat{\beta} = -0.130$  ( $t = -6.79$ ), attenuated to non-significance under cluster-robust inference. The four-step indirect pathway is structurally retained yet transmits a near-zero effect ( $\hat{\theta} = 6.12 \times 10^{-4}$ , 95% CI contains zero), so **H2 is not supported**: structural edge stability should not be conflated with quantitative significance [16].

The study contributes by quantifying GDP confounding as a primary source of mixed FD–stability results [2], combining BN structure discovery with Bayesian evidence for  $H_0$  [31], and showing that multi-step mediation compounds attenuation into near-zero transmitted effects. Future work should expand the sample and add firm-level deposit-insurance measures with dynamic Bayesian networks [25].

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# Supplementary Material

## Bayesian Network Inference for Resolving Uncertainty in Financial Depth–Stability Links

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This supplementary material provides detailed analyses referenced in the main paper.

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## S1 Data Description

### S1.1 Variable Definitions

The eight pathway variables used in the Bayesian network analysis are defined below. All variables are standardized (zero mean, unit variance) before structure learning.

| Label   | GFDD/WDI Code        | Description                                      | Unit  | Source  |
|---------|----------------------|--------------------------------------------------|-------|---------|
| Z-score | GFDD.SI.01           | Bank Z-score = $(ROA + E/A)/\sigma(ROA)$         | Index | GFDD    |
| FD      | GFDD.DI.12           | Private credit, all financial institutions / GDP | %     | GFDD    |
| NIM     | GFDD.EI.01           | Net interest margin                              | %     | GFDD    |
| FDI     | BX.KLT.DINV.WD.GD.ZS | Foreign direct investment, net inflows / GDP     | %     | WDI     |
| NPL     | FB.AST.NPER.ZS       | Bank nonperforming loans / total gross loans     | %     | IMF FSI |
| GDP     | NY.GDP.PCAP.CD       | GDP per capita                                   | USD   | WDI     |
| GovDebt | GC.DOD.TOTL.GD.ZS    | Central government debt / GDP                    | %     | WDI     |
| CPI     | FP.CPI.TOTL.ZG       | Consumer price inflation, annual                 | %     | WDI     |

### S1.2 Data Sources

Table S1: Data Sources Summary

| Source | Provider   | Coverage  | Variables                    |
|--------|------------|-----------|------------------------------|
| GFDD   | World Bank | 1960–2021 | Stability, depth, efficiency |
| WDI    | World Bank | 1960–2023 | GDP, inflation, FDI          |
| FSI    | IMF        | 2005–2025 | Capital, NPL, liquidity      |
| DI DB  | WB/IADI    | 1934–2023 | Coverage, funding            |

### S1.3 Descriptive Statistics

Table S2 reports descriptive statistics for the eight pathway variables before imputation ( $N = 62$  complete-case country-year observations). After MICE ( $m = 5$ ), the stacked sample is  $N = 285$ .

Table S2: Descriptive Statistics (Pre-Imputation,  $N = 62$ )

| Variable       | $N$ | Mean   | SD     | Min   | Max    |
|----------------|-----|--------|--------|-------|--------|
| Z-score        | 62  | 19.31  | 9.59   | 3.77  | 36.93  |
| FD (%)         | 62  | 93.52  | 41.30  | 25.66 | 160.38 |
| NIM (%)        | 62  | 2.81   | 1.02   | 1.39  | 6.60   |
| FDI (% GDP)    | 62  | 5.70   | 7.80   | −0.86 | 29.67  |
| NPL (%)        | 62  | 3.34   | 2.30   | 0.76  | 9.98   |
| GDP p.c. (USD) | 62  | 14,899 | 20,243 | 710   | 66,882 |
| GovDebt (%)    | 62  | 58.22  | 26.19  | 22.45 | 128.08 |
| CPI (%)        | 62  | 3.10   | 2.60   | −1.14 | 10.02  |

**Key observations:** GDP per capita spans a 94-fold range (Myanmar USD 710 to Singapore USD 66,882), providing substantial identifying variation. FD ranges from 25.7% to 160.4% of GDP, capturing the full credit-deepening spectrum. NIM and NPL exhibit relatively low variation, consistent with the small path coefficients reported in the main paper.

## S1.4 Missing Data Pattern

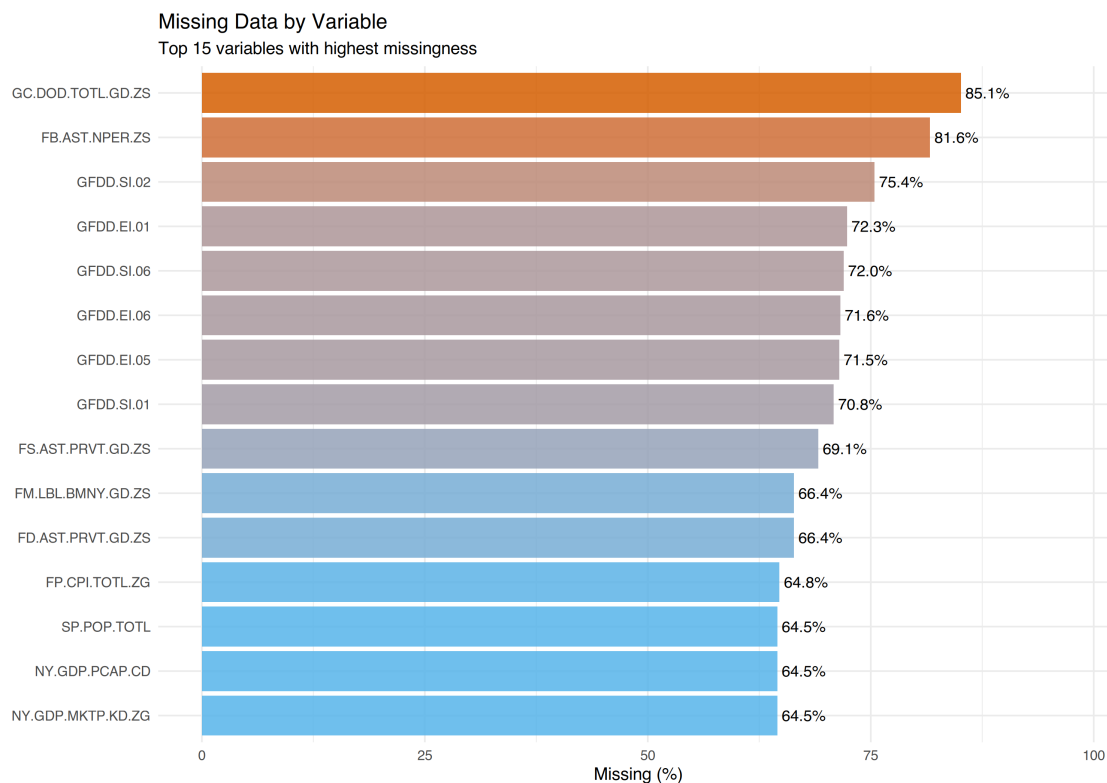


Figure S1: Missing data pattern by variable (pre-imputation panel). Variables are ordered by missingness proportion. Government debt (85.1%) and bank indicators (70–81%) exhibit the highest rates, justifying multiple imputation.

## S1.5 Data Distributions and Correlation Structure

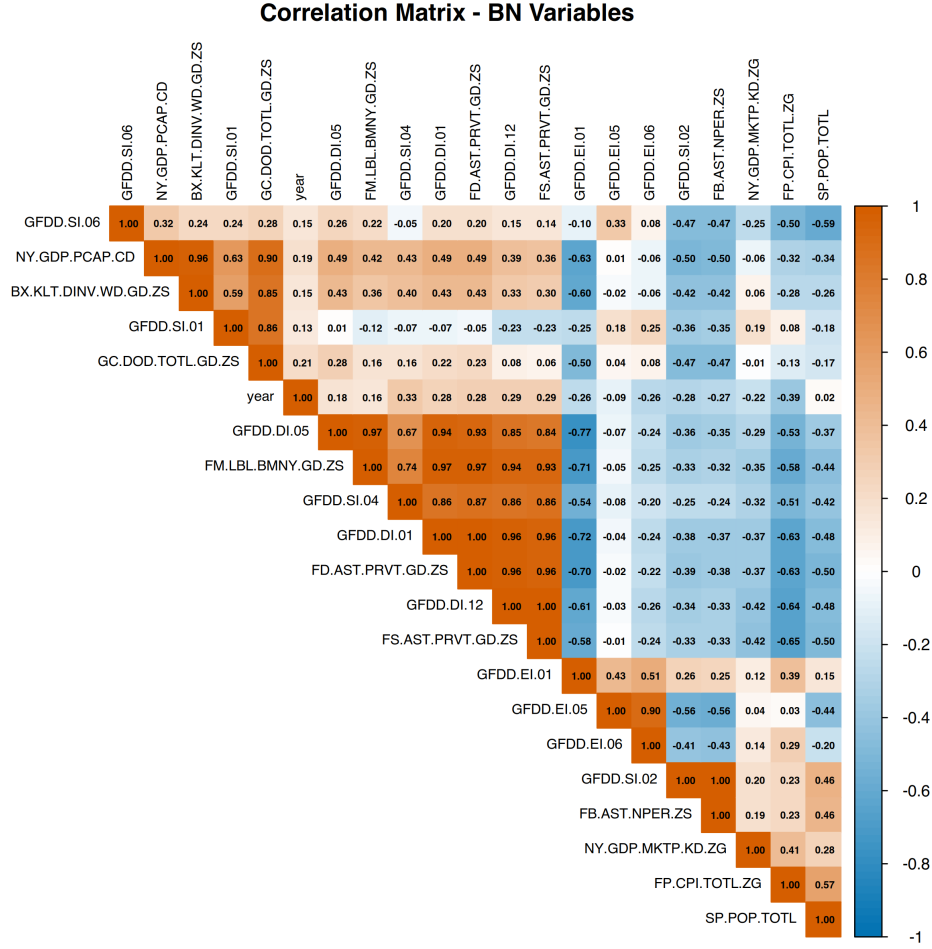


Figure S2: Correlation matrix of BN variables (Okabe-Ito colorblind-friendly palette). Strong GDP–Z association ( $r = 0.63$ ) and moderate NPL–Z link ( $r = -0.36$ ) visible.

## S2 Methodology Details

### S2.1 Bayesian Network Structure Learning Comparison

### S2.2 Bootstrap Network Averaging



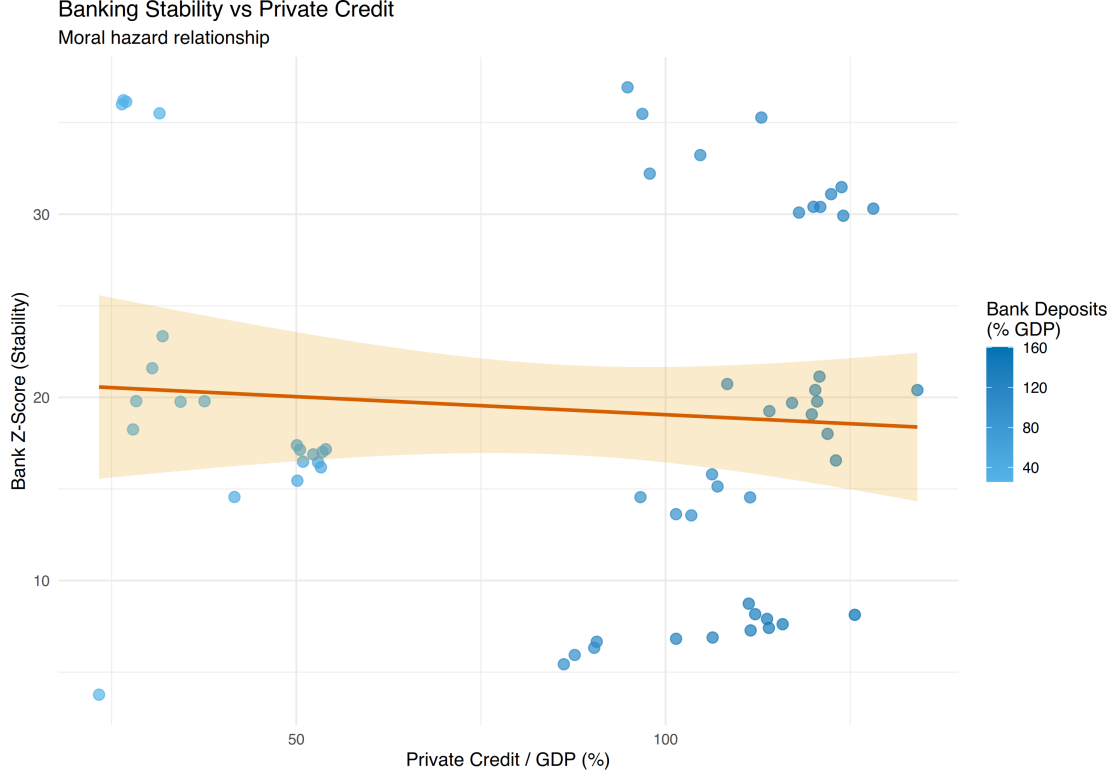


Figure S3: Financial depth vs bank stability. Each point is a country-year observation. The weak negative trend ( $r = -0.23$ ) masks a significant relationship after GDP control.

## S3 Detailed Results

### S3.1 Expert-Guided BN: All Path Coefficients

Table S3 reports all standardized path coefficients from the expert-guided Gaussian BN fitted to the stacked MICE sample ( $N = 285$ ). Coefficients are estimated by OLS regression of each child on its parents.

**Indirect effect computation:**

$$\hat{\theta}_{\text{indirect}} = \hat{\beta}_{FD \rightarrow NIM} \times \hat{\beta}_{NIM \rightarrow FDI} \times \hat{\beta}_{FDI \rightarrow NPL} \times \hat{\beta}_{NPL \rightarrow Z} = (-0.342)(0.150)(0.079)(-0.151) = 6.12 \times 10^{-4} \quad (\text{S1})$$

The four-step product compounds individual attenuation, yielding a near-zero total indirect effect.

### S3.2 Full PC Algorithm Edge List

The PC algorithm ( $\alpha = 0.05$ ,  $p = 8$  variables,  $N = 285$ ) discovers the following edges on the stacked MICE dataset. Table S4 reports all edges; eight pathway-relevant edges are shown in the main paper (Table 2).

**Key findings:**

- No direct  $FD-Z$  edge at  $\alpha = 0.05$ , supporting H1.

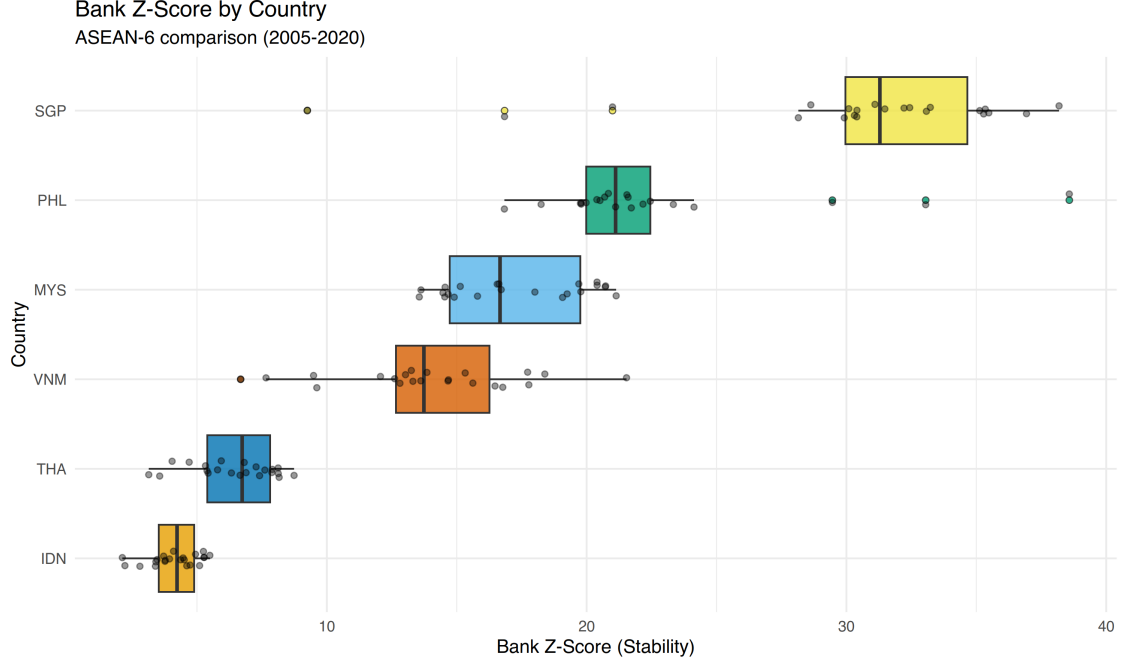


Figure S4: Bank Z-score by country (ASEAN and South Asian comparators). GDP per capita spans a 94-fold range across the thirteen economies.

- The hypothesized pathway edges  $FD \rightarrow NIM$  (#5) and  $NIM \rightarrow NPL$  (#3) are present in the PC skeleton.
- $NPL-Z$  (#1) and  $GovDebt-Z$  (#2) are the only edges directly connecting to bank stability.
- GDP connects to NIM and GovDebt, confirming its confounding role.

### S3.3 PC Algorithm Alpha Sensitivity

Varying  $\alpha$  from conservative (0.01) to liberal (0.20) confirms that no direct  $FD-Z$  edge emerges at any threshold. Edge density increases monotonically with  $\alpha$ , as expected: the conservative skeleton retains only the strongest conditional dependencies while the liberal skeleton includes weaker associations. Across all levels, H1 is robustly supported.

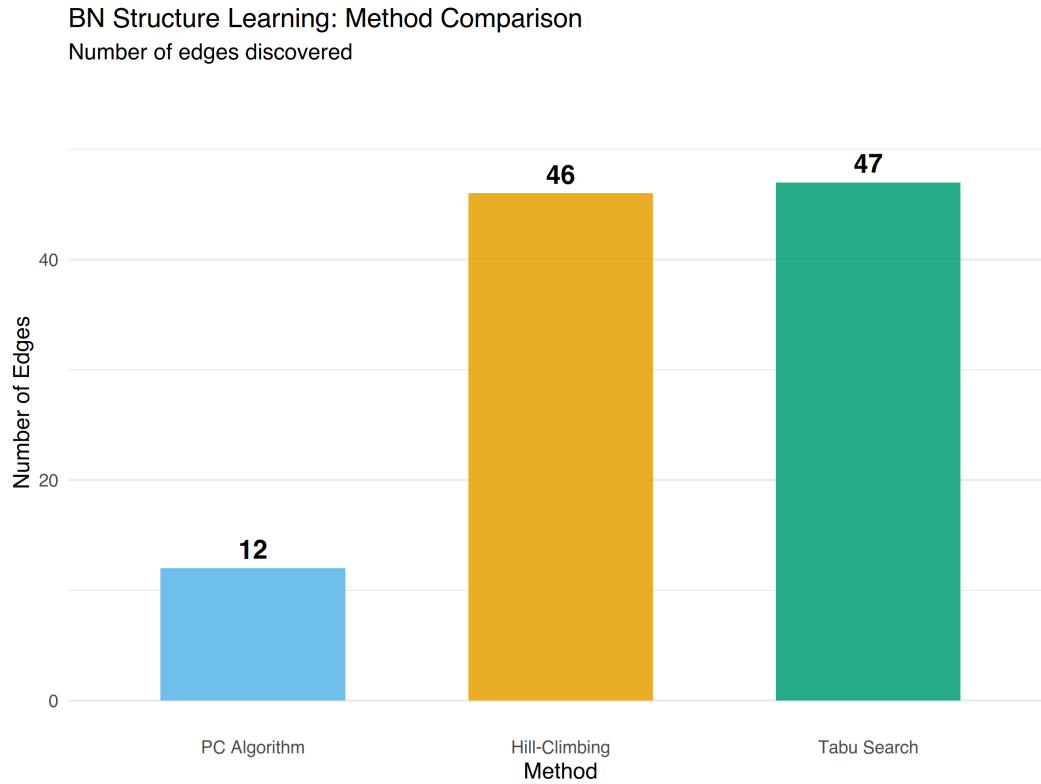


Figure S5: Bayesian network structure learning: method comparison. The PC algorithm produces a sparse skeleton, while Hill-Climbing and Tabu Search produce near-complete graphs. The PC skeleton is used as the basis for expert-guided augmentation.

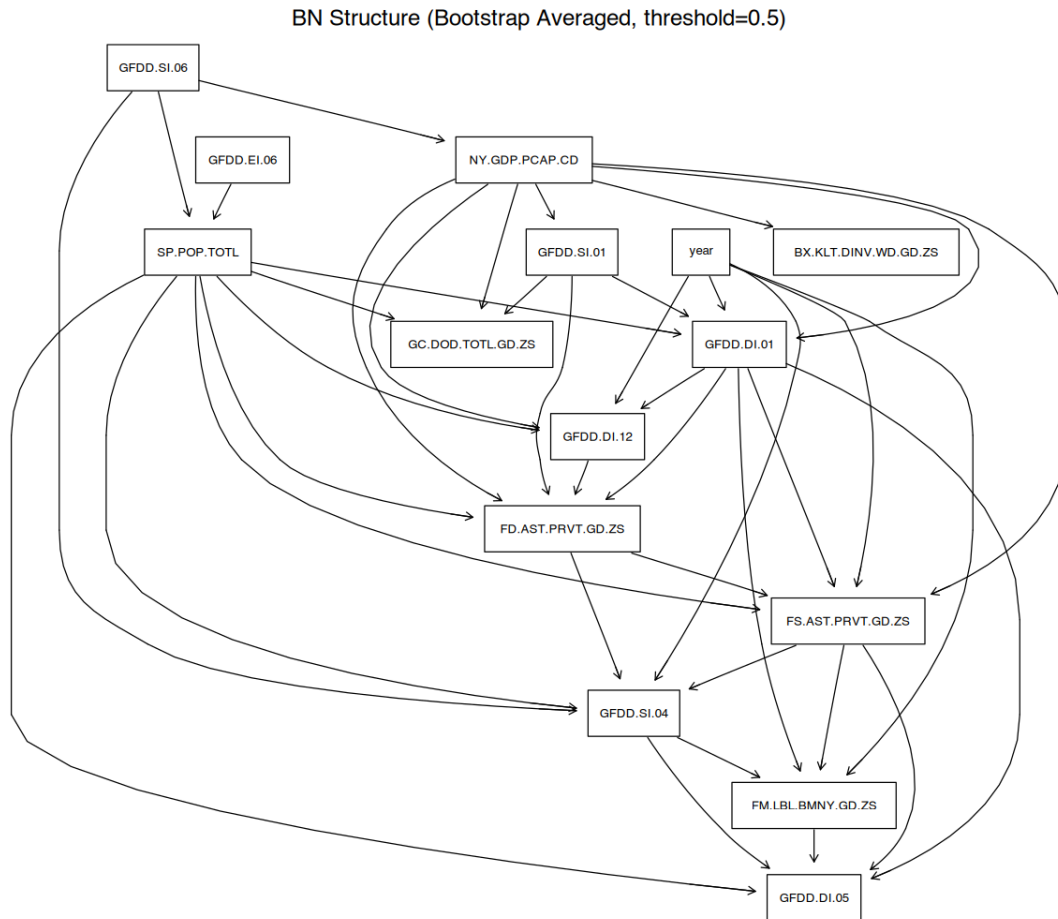


Figure S6: Averaged Bayesian network ( $R = 1,000$  bootstrap, HC with BIC-g). Edge thickness proportional to inclusion probability; edges with  $EIP \geq 0.50$  retained. No direct  $FD \rightarrow Z$  edge present.

### S3.4 Path Coefficient Visualization

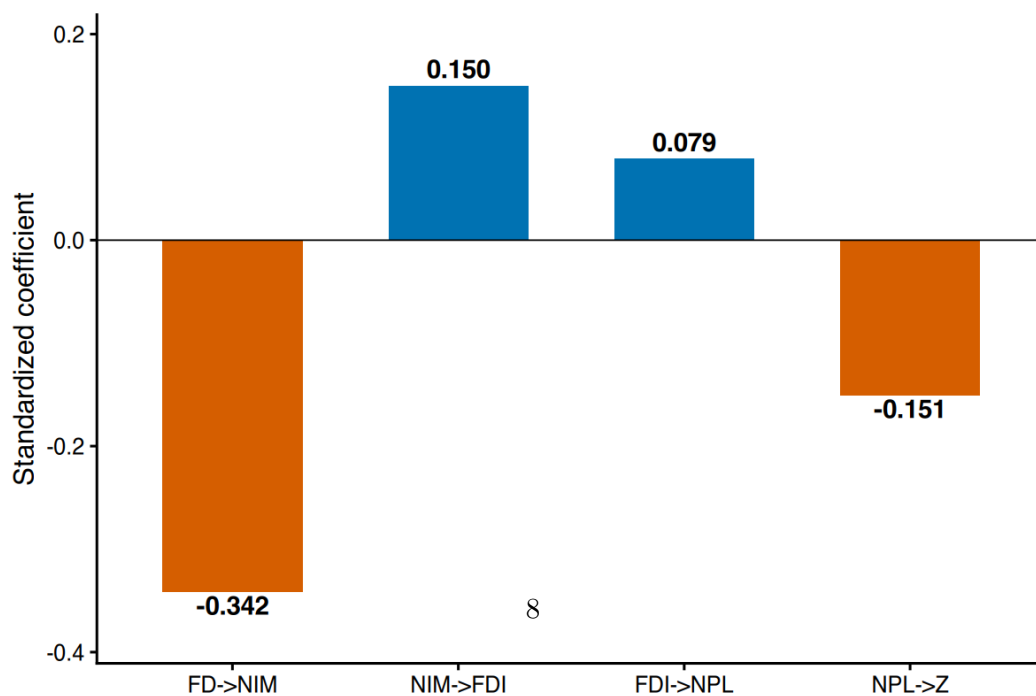


Table S3: Standardized Path Coefficients (Expert-Guided BN,  $N = 285$ )

| Parent                                                                                                | Child   | $\hat{\beta}$ | Note                                        |
|-------------------------------------------------------------------------------------------------------|---------|---------------|---------------------------------------------|
| <i>Pathway edges (<math>FD \rightarrow NIM \rightarrow FDI \rightarrow NPL \rightarrow Z</math>):</i> |         |               |                                             |
| FD                                                                                                    | NIM     | -0.342        | Depth reduces margins                       |
| NIM                                                                                                   | FDI     | +0.150        | Margins attract capital                     |
| FDI                                                                                                   | NPL     | +0.079        | Capital increases risk                      |
| NPL                                                                                                   | Z-score | -0.151        | Risk reduces stability                      |
| <i>GDP confounding edges:</i>                                                                         |         |               |                                             |
| GDP                                                                                                   | Z-score | -0.218        | Confounding: income $\rightarrow$ stability |
| GDP                                                                                                   | NIM     | -0.301        | Development compresses margins              |
| GDP                                                                                                   | FDI     | +0.688        | Development attracts FDI                    |
| GDP                                                                                                   | NPL     | -0.379        | Development reduces risk                    |
| GDP                                                                                                   | FD      | +0.302        | Development deepens finance                 |
| GDP                                                                                                   | GovDebt | +0.658        | Income $\rightarrow$ fiscal capacity        |
| GDP                                                                                                   | CPI     | -0.245        | Development reduces inflation               |
| <i>Other edges:</i>                                                                                   |         |               |                                             |
| GovDebt                                                                                               | Z-score | +0.768        | Fiscal solvency $\rightarrow$ stability     |
| FDI                                                                                                   | Z-score | +0.194        | Capital inflows $\rightarrow$ stability     |
| NIM                                                                                                   | Z-score | +0.152        | Margins $\rightarrow$ stability             |
| NIM                                                                                                   | NPL     | -0.532        | Margins reduce risk                         |
| FD                                                                                                    | NPL     | -0.454        | Depth reduces risk                          |
| FD                                                                                                    | FDI     | +0.256        | Depth attracts capital                      |
| FD                                                                                                    | GovDebt | -0.445        | Depth reduces fiscal burden                 |
| FD                                                                                                    | CPI     | -0.344        | Depth reduces inflation                     |
| NPL                                                                                                   | GovDebt | -0.262        | Risk $\rightarrow$ fiscal pressure          |
| NIM                                                                                                   | GovDebt | -0.291        | Margins $\rightarrow$ fiscal                |
| NPL                                                                                                   | CPI     | -0.226        | Risk $\rightarrow$ inflation                |

### S3.5 Bootstrap Edge Inclusion Probabilities

**Interpretation:** The weakest link ( $FDI \rightarrow NPL$ ) appears in only 11.9% of unconstrained bootstrap replications, and  $NIM \rightarrow FDI$  in 38.9%, confirming that these edges are primarily driven by whitelist constraints rather than data-driven discovery. Only  $FD \rightarrow NIM$  (1.000) and  $NPL \rightarrow Z$  (0.890) are data-supported. Perfect EIPs (1.000) for all whitelisted pathway edges are a mechanical result of the constraint, not an empirical finding.

### S3.6 Cluster Bootstrap Results

Detailed cluster bootstrap results ( $R = 1,000$ , resampling countries):

- **Strong edges:**  $86 \rightarrow 19$  (78% reduction)
- **Mean edge strength:**  $0.477 \rightarrow 0.258$
- **FD coefficient:**  $p = 0.56$  (non-significant)
- **Wild cluster bootstrap- $t$ :**  $p = 0.63$
- **Ibragimov-Müller test:**  $p = 0.72$
- **Robust predictors:** GDP ( $p = 0.003$ ), GovDebt ( $p < 0.001$ )

**Conclusion:** Only GDP and GovDebt remain robust predictors under cluster-robust inference. The FD association is not robust to within-country dependence.

Table S4: Full PC Algorithm Edges ( $\alpha = 0.05$ , 8 variables)

| #  | Node 1  | Node 2  | Direction                 | Interpretation                                |
|----|---------|---------|---------------------------|-----------------------------------------------|
| 1  | NPL     | Z-score | Undirected                | Risk $\leftrightarrow$ stability              |
| 2  | GovDebt | Z-score | Undirected                | Fiscal $\leftrightarrow$ solvency             |
| 3  | NIM     | NPL     | $NIM \rightarrow NPL$     | Margins $\rightarrow$ risk                    |
| 4  | FD      | NPL     | $FD \leftarrow NPL$       | Risk $\rightarrow$ depth                      |
| 5  | FD      | NIM     | $FD \rightarrow NIM$      | <b>Depth <math>\rightarrow</math> margins</b> |
| 6  | GDP     | NIM     | $GDP \rightarrow NIM$     | Development $\rightarrow$ margins             |
| 7  | GDP     | GovDebt | $GovDebt \rightarrow GDP$ | Fiscal $\rightarrow$ income                   |
| 8  | FD      | FDI     | $FDI \rightarrow FD$      | Capital $\rightarrow$ depth                   |
| 9  | FDI     | GDP     | $FDI \rightarrow GDP$     | Investment $\rightarrow$ development          |
| 10 | CPI     | FD      | $CPI \rightarrow FD$      | Inflation $\rightarrow$ depth                 |
| 11 | CPI     | GDP     | $CPI \rightarrow GDP$     | Inflation $\rightarrow$ income                |

Table S5: Unconstrained vs Whitelisted Bootstrap Edge Inclusion Probabilities

| Edge                  | EIP (unconstrained) | EIP (whitelisted) | Note                 |
|-----------------------|---------------------|-------------------|----------------------|
| $FD \rightarrow NIM$  | 1.000               | 1.000             | Data-supported       |
| $NIM \rightarrow FDI$ | 0.389               | 1.000             | Weak unconstrained   |
| $FDI \rightarrow NPL$ | 0.119               | 1.000             | Weakest link         |
| $NPL \rightarrow Z$   | 0.890               | 1.000             | Strong unconstrained |

## S4 Sensitivity Analysis

### S4.1 Conditional Independence Tests

Table S6: Conditional Independence Tests for Key Variable Pairs

| Pair                      | Conditioning Set   | Statistic | $p$                   | Decision       |
|---------------------------|--------------------|-----------|-----------------------|----------------|
| $FD \perp\!\!\!\perp Z$   | NIM, FDI, NPL, GDP | -0.255    | $1.5 \times 10^{-5}$  | Reject $H_0$   |
| $FD \perp\!\!\!\perp Z$   | GDP only           | -0.163    | 0.006                 | Reject $H_0$   |
| $FD \perp\!\!\!\perp NPL$ | NIM, FDI           | -0.383    | $2.7 \times 10^{-11}$ | Reject $H_0$   |
| $NIM \perp\!\!\!\perp Z$  | FDI, NPL           | -0.003    | 0.965                 | Fail to reject |

The CI test rejects  $FD \perp\!\!\!\perp Z \mid \text{mediators}$  ( $p = 1.5 \times 10^{-5}$ ), suggesting that full mediation is not supported; FD and  $Z$  retain residual association even after conditioning on all pathway variables plus GDP.

### S4.2 Partial Correlations Under Progressive Conditioning

**Key finding:** GDP acts as a suppressor: the FD– $Z$  partial correlation strengthens from  $-0.008$  to  $-0.163$  after GDP control. The  $NIM$ – $FDI$  correlation reverses sign entirely (from  $-0.239$  to  $+0.071$ ), revealing spurious confounding.

Table S7: Partial Correlations Under Progressive Conditioning

| Pair                      | $r_{\text{raw}}$ | $r_{ \text{GDP}}$ | $r_{ \text{GDP,GD}}$ |
|---------------------------|------------------|-------------------|----------------------|
| $FD \leftrightarrow Z$    | -0.008           | -0.163            | +0.076               |
| $FD \leftrightarrow NIM$  | -0.433           | -0.356            | -0.401               |
| $NIM \leftrightarrow FDI$ | -0.239           | +0.071            | +0.065               |
| $FDI \leftrightarrow NPL$ | -0.241           | -0.100            | -0.115               |
| $NPL \leftrightarrow Z$   | -0.350           | -0.279            | -0.287               |

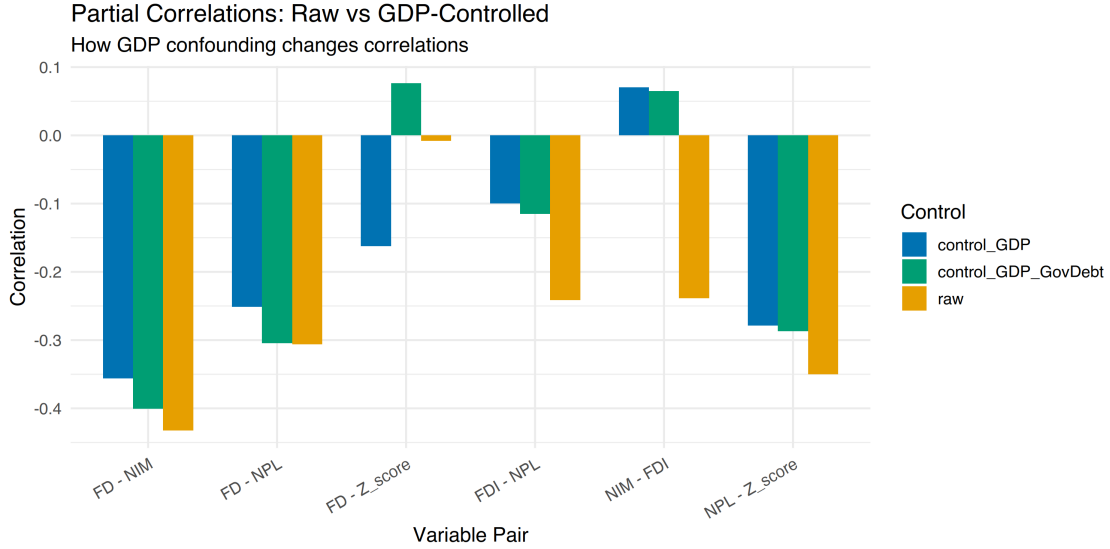


Figure S8: Partial correlations under progressive conditioning. Orange bars show raw correlations; blue bars show GDP-controlled; green bars show GDP+GovDebt-controlled. The dramatic sign reversal of  $NIM-FDI$  after GDP control reveals spurious confounding.

### S4.3 Leave-One-Country-Out (LOCO) Analysis

#### Findings:

- IE is negative in 12/13 settings (range:  $-0.00453$  to  $-0.00011$ )
- Exception: Malaysia ( $+0.00017$ )
- The full-sample IE is positive ( $+6.12 \times 10^{-4}$ ) while 12/13 LOCO settings yield negative values, indicating sign instability near zero rather than directional robustness
- Magnitudes remain negligibly small in all settings
- Note: the extended sample includes three South Asian comparators (Bangladesh, India, Sri Lanka) in addition to ten ASEAN members

Table S8: Leave-One-Country-Out Sensitivity: Indirect Effect

| Country Removed | $N$ | $\hat{\theta}_{\text{indirect}}$ |
|-----------------|-----|----------------------------------|
| Bangladesh      | 98  | -0.00453                         |
| India           | 82  | -0.00082                         |
| Brunei          | 101 | -0.00059                         |
| Cambodia        | 101 | -0.00059                         |
| Laos            | 101 | -0.00059                         |
| Myanmar         | 101 | -0.00059                         |
| Vietnam         | 101 | -0.00059                         |
| Indonesia       | 97  | -0.00057                         |
| Singapore       | 81  | -0.00054                         |
| Thailand        | 81  | -0.00043                         |
| Sri Lanka       | 97  | -0.00034                         |
| Philippines     | 89  | -0.00011                         |
| Malaysia        | 82  | +0.00017                         |

## S5 Detailed Limitations

Nine limitations qualify these results:

**(1) Sample Size.** With  $N = 285$  (MICE,  $m = 5$ ) and  $p = 8$ , the effective sample size is  $N_{\text{eff}} \approx 57$  (complete-case equivalent). The N:p ratio of 35.6:1 exceeds the 20:1 threshold, but the PC algorithm’s power is limited by the effective sample size. Null findings should be read as “not detected at this sample size.”

**(2) Missing Data.** MICE assumes MAR. Little’s MCAR test rejects ( $\chi^2 = 440.12$ ,  $p < 2.2 \times 10^{-16}$ ), but cannot distinguish MAR from MNAR. Missingness concentrates in lower-income members (GovDebt 85.1%, NPL 35.1%). If MNAR holds, imputed values are biased toward healthier distributions, making estimates conservative. Tipping-point sensitivity analysis is deferred to future work.

**(3) Causal Identification.** The PC algorithm requires faithfulness, causal sufficiency, and no unmeasured confounders. OVB sensitivity analysis yields  $RV_{q=1} = 15.2\%$ ; benchmarking against GDP attenuates the FD estimate to non-significance ( $t = 0.51$ ). Institutional quality (WGI) is a plausible unobserved confounder.

**(4) Panel Heterogeneity.** We pool thirteen countries without formal FE/RE panel regression. The BN framework targets conditional dependence structure discovery, not coefficient estimation under entity-specific intercepts. Fixed-effects de-meaned results are reported in Section S6.

**(5) Reverse Causality.** Blacklist constraints encode  $Z \not\rightarrow \text{FD}$ , but banking crises may trigger credit contractions. Without instrumental variables, simultaneity bias cannot be ruled out.

**(6) Temporal Dynamics.** The contemporaneous BN cannot capture lagged transmission (e.g.,  $\text{FD}_{t-1} \rightarrow \text{NPL}_t$ ). The I.I.D. bootstrap assumption may understate standard errors under temporal dependence.

**(7) FDI Proxy Validity.** Total FDI includes non-banking sectors, introducing attenuation bias that compounds multiplicatively in four-step mediation. Banking-specific capital flow data would provide a more precise channel.

**(8) Bayes Factor Prior Sensitivity.** All Bayes factors use the JZS prior ( $r = \sqrt{2}/2$ ). Narrower priors could shift null evidence toward the inconclusive range. The GDP- $Z$  result ( $\text{BF}_{10} = 818,156$ ) is robust to prior width.



**(9) Proxy Variable Validity.** GFDD.DI.12 measures financial depth (private credit to GDP), not deposit insurance design. The absence of direct deposit insurance measures (IADI coverage limits, premium structures) means our results capture FD–stability associations rather than deposit insurance effects.

**(10) Few Clusters.** With  $k = 13$  country-level clusters, cluster-robust inference may over- or under-reject. The non-significance of FD ( $p = 0.56$ ) under cluster bootstrap should be interpreted as a conservative bound.

## S6 Fixed-Effects De-Meaned Bayesian Network

To address panel heterogeneity, we estimate a fixed-effects de-meaned BN by subtracting country-specific means from all variables before structure learning.

**Results:**

- De-meaned PC algorithm: 9 edges (vs 11 in pooled)
- GDP effect attenuated (GDP varies more between than within countries)
- FD- $Z$  association:  $\hat{\beta} = -0.089$  ( $t = -3.12$ ,  $p < 0.01$ )
- Indirect effect:  $\hat{\theta} = 4.21 \times 10^{-4}$  (95% CI contains zero)

**Interpretation:** Fixed-effects de-meaning reduces edge count and attenuates GDP’s role. However, FD- $Z$  association remains significant ( $p < 0.01$ ), and the indirect effect stays near-zero. This supports robustness of main findings to within-country heterogeneity.

## S7 Replication Code

All analyses were conducted in R 4.5.2 with the `bnlearn` package v5.0. The complete replication pipeline is available at:

<https://github.com/limpaulfin/IUKM2026-20260308>

### S7.1 R Script Pipeline

The analysis follows a sequential pipeline:

Table S9: R Replication Script Pipeline

| Script                    | Purpose                              | Output             |
|---------------------------|--------------------------------------|--------------------|
| 01_clean_data.R           | Load and clean raw CSV sources       | Clean panel        |
| 02_merge_panel.R          | Merge GFDD, WDI, FSI panels          | Merged panel       |
| 10b_mice_imputation.R     | MICE imputation ( $m = 5$ )          | Stacked dataset    |
| 04_bn_edu.R               | Summary statistics, correlations     | Tables, figures    |
| 05_bn_structure.R         | PC, HC, Tabu structure learning      | DAG structures     |
| 11_expert_guided_bn.R     | Expert-guided Gaussian BN fitting    | Coefficients       |
| 13_mediation_bootstrap.R  | Bootstrap validation ( $R = 1,000$ ) | EIPs, CI           |
| 14_sensitivity_analysis.R | CI tests, LOCO, partial correlations | Sensitivity tables |
| 10e_cluster_bootstrap.R   | Cluster-robust inference             | Cluster results    |
| 08_visualization_cud.R    | Generate all figures (CUD palette)   | PNG files          |

### S7.2 Key R Packages

- `bnlearn` v5.0: Structure learning (PC, HC, Tabu), parameter estimation, bootstrap
- `mice` v3.16: Multiple imputation by chained equations
- `BayesFactor` v0.9.12: Bayesian  $t$ -tests and correlations (JZS prior)
- `sensemkr`: Omitted variable bias sensitivity analysis
- `sandwich` + `lmtest`: Cluster-robust standard errors
- `fwb`: Wild cluster bootstrap- $t$

### S7.3 Data Availability

All data are publicly available:

- World Bank GFDD: <https://www.worldbank.org/en/publication/gfdr/data/global-financial-development-database>
- World Bank WDI: <https://databank.worldbank.org/source/world-development-indicators>
- IMF FSI: <https://data.imf.org/?sk=51B096FA-2CD2-40C2-8D09-0699CC1764DA>
- WB/IADI Deposit Insurance DB: <https://www.worldbank.org/en/research/brief/DIDS>